

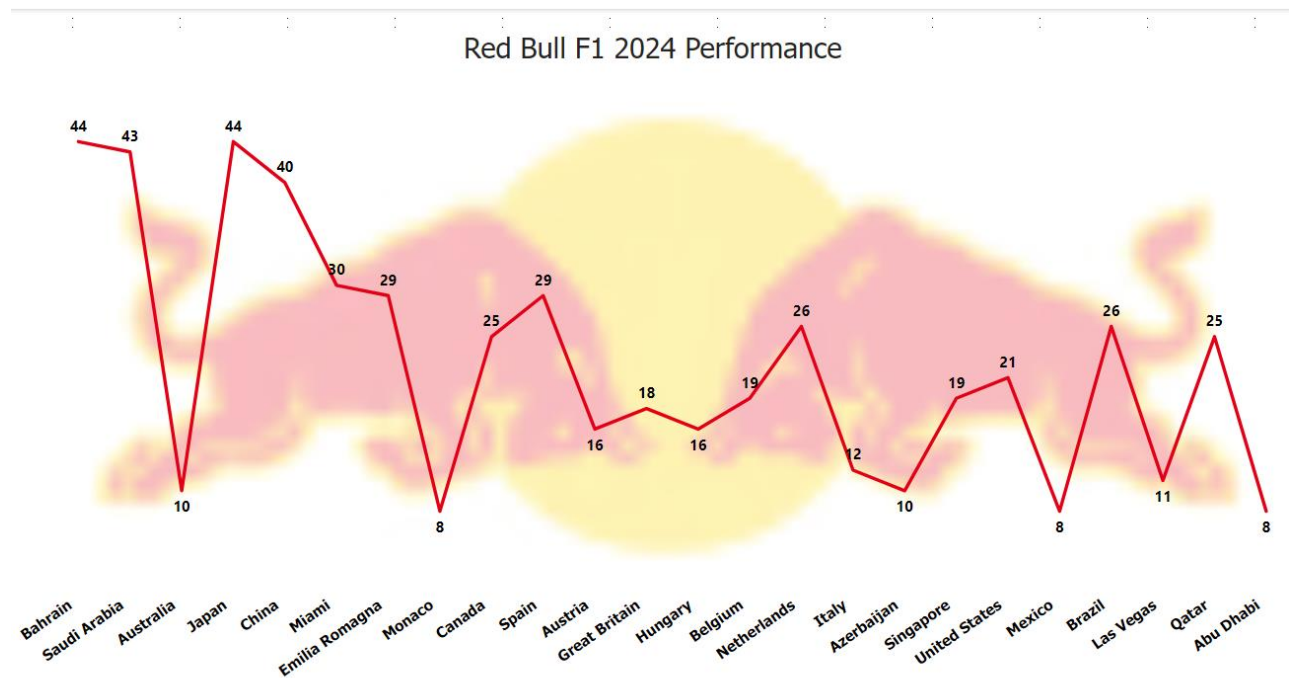
Red Bull Racing: From Dominance to Decline.

The 2024 Formula 1 season was a tale of highs and lows for Red Bull Racing Honda RBT.

Following a record-breaking 2023 season, where they secured the Constructors' Championship with 760 points, Red Bull entered 2024 with high expectations.

Max Verstappen continued to showcase his brilliance, but the team's performance as a whole faltered, leading to a third-place finish with 589 points, 171 points behind their 2023 points tally.

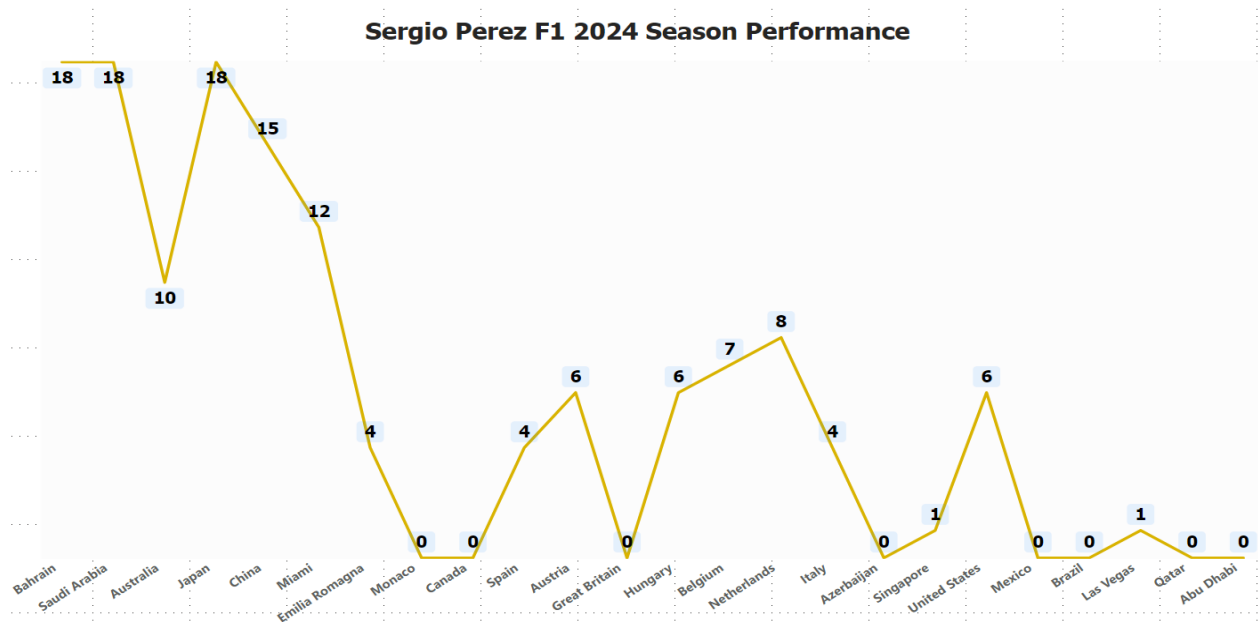
This was a significant drop from their previous dominance, where both Verstappen and Sergio Pérez occupied the top two spots in the Drivers' Championship.



Red Bull began the season strong, dominating the opening races and leaving their competitors trailing. However, as the season progressed, cracks began to show. The mid-season marked a turning point, with a noticeable decline in car performance and inconsistent results from Sergio Pérez.

Performance Decline

Red Bull's struggles raised questions about the role of team strategies and driver performance in shaping their season.



Sergio Pérez's contribution to the team's struggles was significant. His performance dipped dramatically, towards the mid-season.

Early in the season, Pérez seemed poised to replicate his 2023 success, but his points tally took a sharp downturn starting in Miami and struggled to rise there on after.

Towards the end of the season, he barely had any points and ended it on a DNF in Abu Dhabi.

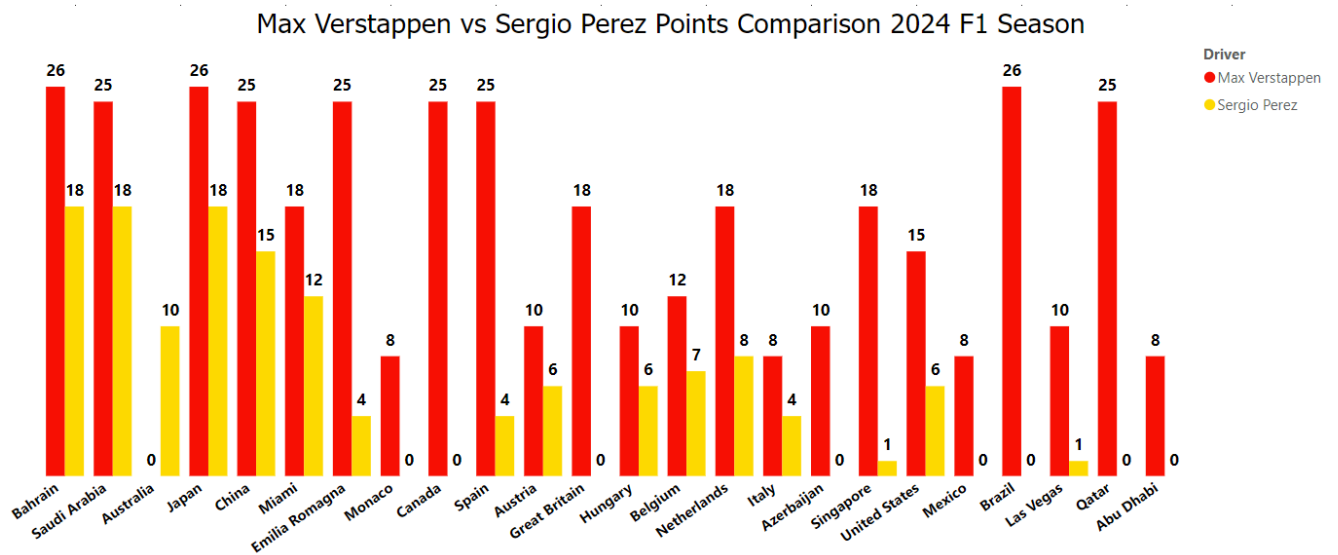
DNFs
5

The issues were compounded by five DNFs in Monaco, Canda, Azerbaijan, Qatar and Abu Dhabi.

In Great Britain, Mexico and Brazil he finished outside points totaling 8 races with no points which did nothing to help Red Bull's fight for the constructor championships.

Races without Points
8

Max vs Perez



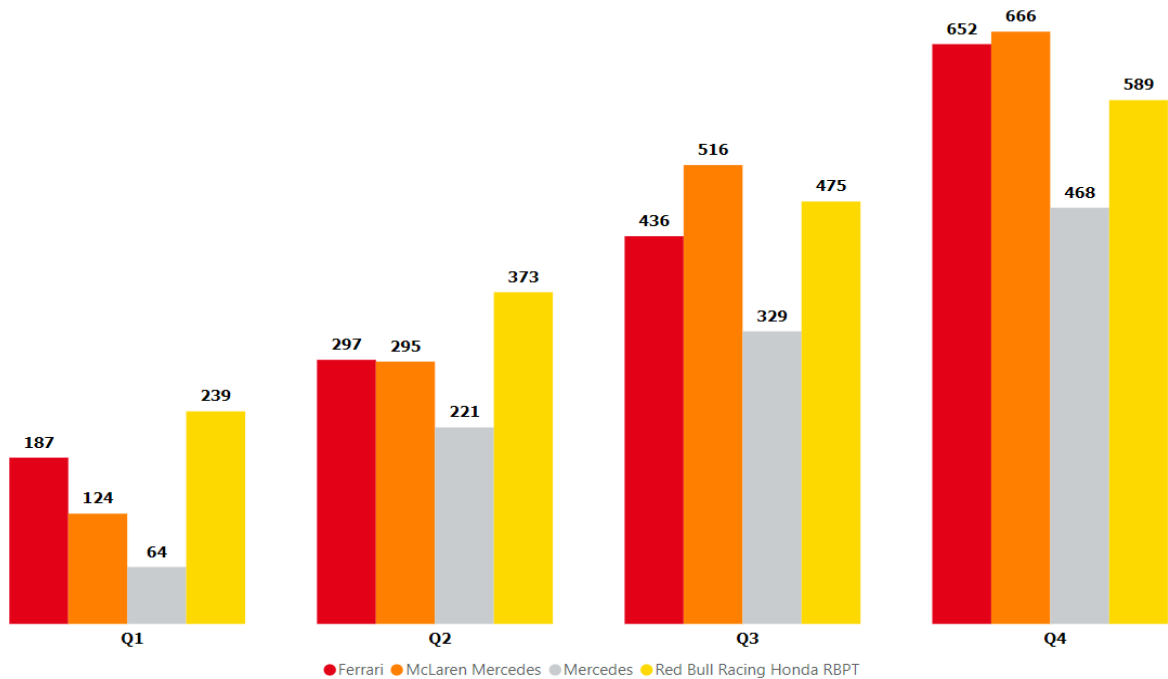
When compared to Max Verstappen, the disparity in their performances becomes stark. Verstappen consistently delivered podium finishes, while Pérez's inability to score crucial points hindered Red Bull's championship hopes. This gap showed just how much a team's performance can suffer when their second driver falls short.

The red bull car struggled during the mid-season which is evident in the points dip of both drivers starting in Austria. Despite that, Max held his own and managed to deliver some points, even rarely straying from double digits. Perez on the other hand declined steadily in his race performance and barely had points towards the end of the F1 2024 season.

Competitor Success

While Red Bull grappled with internal challenges, rivals McLaren, Mercedes, and Ferrari seized the opportunity to gain ground. McLaren, led by the exceptional performances of Lando Norris and Oscar Piastri, showcased a remarkable resurgence. Mercedes and Ferrari also capitalized on Red Bull's struggles, often splitting points that would otherwise have been Red Bull's.

Top 4 Teams Cummulative Points by Quarters F1 2024 Season

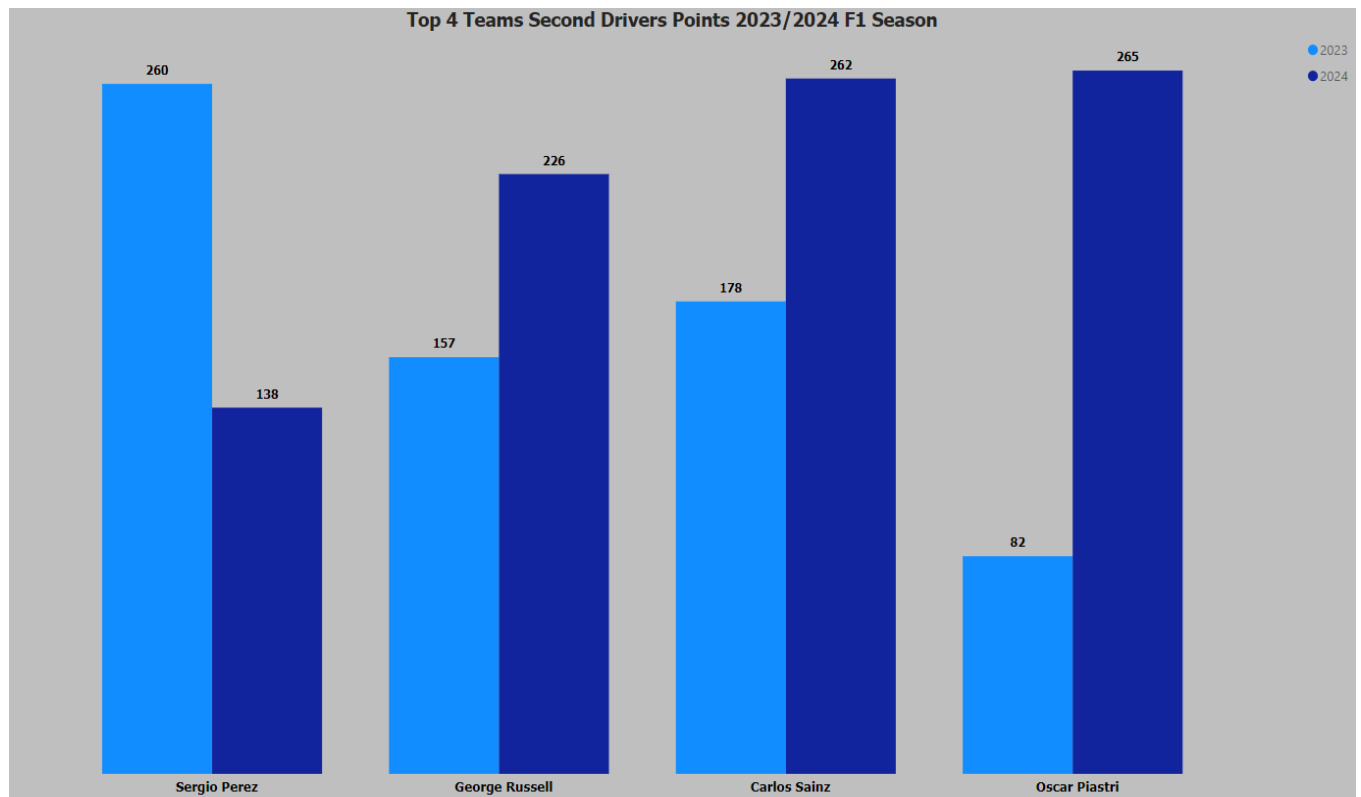


During the first quarter, Red Bull maintained a commanding lead. By the second quarter, McLaren and Ferrari had started catching up, and by the third quarter, McLaren established a significant lead, positioning themselves as strong contenders for the Constructors' Championship.

The rest of the season saw McLaren, Mercedes, and Ferrari closing the gap and outperforming Red Bull. This momentum shift highlights how crucial consistent performance is in maintaining championship dominance.

Comparing Second Drivers

Taking a closer look at the second drivers from the top four teams highlights just how much Pérez's underperformance affected Red Bull's season.



Comparing the cumulative points of second drivers of the Top 4 teams in 2023 and 2024 paints a clear picture of how poor Pérez's performance was in the 2024 F1 season compared to his past performances.

While the other teams' second drivers improved or maintained their form, Pérez's performance declined drastically.

With a difference of 122 points in his points tally in the 2023 F1 season and the 2024 F1 season, Sergio was well behind the other Top 4 second drivers.



In 2023, Pérez's points tally (260) was on par with other second drivers. This contributed significantly to Red Bull's success in winning the 2023 Constructors' championship.

However, in 2024, he fell behind, with his cumulative points (138) well below the average point tally of the second drivers from McLaren, Mercedes, and Ferrari.

Conclusion

Red Bull's 2024 season was a stark reminder of the importance of a balanced team.

The significant gap in average cumulative points among second drivers in 2024 shows a glaring disparity between Pérez and his counterparts.

Had Red Bull replaced Pérez with a driver performing at the level of other top second drivers, the team could have amassed significantly more points and potentially finished higher in the championship standings.

In retrospect, Red Bull's decision to retain Pérez for the entire season might have cost them crucial points.

The combination of declining car performance, inconsistent driver results, and strong competition from McLaren, Mercedes, and Ferrari proved too much to overcome. Moving forward, Red Bull will need to address these challenges to reclaim their dominance and ensure a more competitive lineup in future seasons.

Critical Reflection

For this coursework, I used data from the 2023 and 2024 Formula 1 data sets, which I found on a public GitHub repository (<https://github.com/toUpperCase78/formula1-datasets>).

I chose these datasets because they were accurate and had all the details I needed to answer my research question: "Would Red Bull Racing have secured a stronger championship finish if Sergio Pérez had been replaced earlier in the 2024 season?" I looked through public data websites like Kaggle and GitHub to find the best data.

The datasets had detailed stats for each race, including lap times and race points, which were important for my analysis. The data was up-to-date and factually correct, making it reliable for my coursework.

The sprint results data were separate from the race results data, which made it tricky to calculate cumulative points and there was some missing data for Logan Sargeant in the Austria Grand Prix, which needed extra verification. Handling the missing value for Logan Sargeant was a bit of a challenge, but it was the only major data gap.

I used Excel to clean the data and handle any missing values. To ensure accuracy, I fact-checked the data using the official Formula 1 website. I successfully transformed the data using Power Query in Power BI, creating the necessary visualizations to tell my data story.

The visualizations were formatted using team colors and appropriate formatting, which helped in presenting a coherent and visually appealing data story. The visualizations effectively highlighted key insights, such as the performance differences between Sergio Pérez, Max Verstappen, his teammate, and other drivers.

By using team colors and consistent formatting I was able to create a visually appealing and coherent narrative.

I would suggest that adding predictive analysis could provide deeper insights into potential outcomes if Red Bull had made different decisions earlier in the season.

I would have also liked to compare Sergio Pérez to all drivers, not just the top four, as this would give a more comprehensive view of the 2024 season, reflecting the growth and improvement of other teams.

In conclusion, I successfully used detailed and accurate datasets to address the research question, and the data preparation and visualization processes were well-executed, although there is room for improvement in expanding the scope of comparisons and incorporating predictive analysis.